Lab 35 Worksheet — Linux Permissions, Ownership and User Management

Name: ____ **Date:** __

Exam objective: Apply the Linux permission model to users, groups and others using symbolic and numeric notation, and manage users and sudo rights (Core 2 objective 1.11).

Record as you go

Step	What you did	What you observed
1	Open the Killercoda playground and examine a file's permission string, identifying the three permission triplets for owner, group and others.	
2	Record the notation: r equals 4, w equals 2, x equals 1, so rwx is 7, rw- is 6, r-x is 5 and r-- is 4.	
3	Build the conversion table for the modes you will actually see: 777, 755, 750, 700, 644, 640 and 600, giving the symbolic form and the typical use of each.	
4	Set permissions numerically and verify the symbolic result matches your table.	
5	Set permissions symbolically and confirm it produces the same result as the numeric form.	
6	Record why directories need the execute bit: without x on a directory you cannot enter it or access anything inside, even with read permission.	
7	Create two users to demonstrate real access control between accounts.	
8	Create a shared group and add both users to it.	
9	Create a shared directory owned by the group with permissions that allow the group to write but	

Step	What you did	What you observed
	exclude everyone else.	
10	Set the setgid bit so that files created inside inherit the group, which is what makes shared directories actually work in practice.	
11	Verify the access control works: alice can create a file, and the file inherits the support group from the setgid bit.	
12	Grant alice administrative rights through sudo and record why sudo is preferred over logging in as root.	

Verification

Success criterion: Your conversion table is correct for all seven modes; the execute-bit demonstration shows a directory becoming enterable at 755 but not at 644; /srv/shared is mode 2770 owned by the support group; and alice's file inherits the support group.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
